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DISPENSER ASSEMBLY INCLUDING AN OPEN CHANNEL LINE PURGE

Abstract

A method of operating a dispenser assembly of an appliance includes receiving a signal to dispense a predetermined volume of a fluid from at least one fluid storage vessel via a fluid diverter; selecting a first fluid inlet channel in response to receiving the signal, the first fluid inlet channel being fluidly connected with the at least one fluid storage vessel; initiating a pump to pump the predetermined volume of the fluid from the at least one fluid storage vessel after selecting the first fluid inlet channel; and selecting a second fluid inlet channel of the plurality of fluid inlet channels after pumping the predetermined volume of the fluid, the second fluid inlet channel being an open-air inlet channel.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present subject matter relates generally to domestic appliances, and more particularly to dispenser assemblies for dishwasher appliances.

BACKGROUND OF THE INVENTION

[0002] Certain domestic appliances, such as dishwasher appliances, generally include a tub that defines a wash chamber and a pump for directing fluids from a fluid reservoir into the wash chamber during washing operations. A detergent dispenser for dispensing detergent into the wash chamber may be included. In some instances, the detergent dispenser is configured to dispense detergent or other fluids from one or more bulk tanks provided within the appliance. Such dispensers may either mix the fluids or maintain distinct channels for each respective bulk tank, depending on which fluid or detergent is to be added to the wash chamber.

[0003] Mixing the fluids is undesirable in some cases, such a dishwasher cycle where specific liquids need to be dispensed at specific times. Thus, when a first fluid is to be dispensed at a first time and a second fluid is to be dispensed at a second time, the fluids may be mixed within a common flow tube. Further, maintaining distinct channels can be undesirable because separate control devices are required for all channels, thereby adding complexity.

[0004] Accordingly, a dispensing assembly for an appliance that obviates one or more of the above-mentioned drawbacks would be beneficial. In particular, a dispensing assembly including features for purging a common flow tube between dispensations would be useful.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

[0006] In one exemplary aspect of the present disclosure, a dispenser assembly for an appliance is provided. The dispenser assembly may include a fluid diverter including a plurality of fluid inlet channels, a fluid outlet channel, and a diverter input shaft; a pump fluidly coupled to the fluid diverter for selectively urging a flow of fluid from the fluid diverter into the appliance; at least one fluid storage vessel configured to store a fluid, the at least one fluid storage vessel fluidly connected with the fluid diverter via a first fluid inlet channel of the plurality of fluid inlet channels; and a controller operably connected with the fluid diverter and the pump, the controller being configured to perform an operation. The operation may include receiving a signal to dispense a predetermined volume of the fluid from the at least one fluid storage vessel via the fluid diverter; selecting the first fluid inlet channel in response to receiving the signal; initiating the pump to pump the predetermined volume of the fluid from the at least one fluid storage vessel after selecting the first fluid inlet channel; and selecting a second fluid inlet channel of the plurality of fluid inlet channels after pumping the predetermined volume of the fluid, wherein the second fluid inlet channel is an open-air inlet channel.

[0007] In another exemplary aspect of the present disclosure, a method of operating a dispenser assembly of an appliance is provided. The dispenser assembly may include a fluid diverter including a plurality of fluid inlet channels and a fluid outlet channel, a pump fluidly coupled to the fluid diverter, and at least one fluid storage vessel. The method may include receiving a signal to dispense a predetermined volume of a fluid from the at least one fluid storage vessel via the fluid diverter; selecting a first fluid inlet channel in response to receiving the signal, the first fluid inlet channel being fluidly connected with the at least one fluid storage vessel; initiating the pump to pump the predetermined volume of the fluid from the at least one fluid storage vessel after selecting the first fluid inlet channel; and selecting a second fluid inlet channel of the plurality of fluid inlet channels after pumping the predetermined volume of the fluid, wherein the second fluid inlet channel is an open-air inlet channel.

[0008] These and other features, aspects and advantages of the present invention will become better

understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

[0010] FIG. 1 provides a front view of a dishwashing appliance in accordance with an exemplary embodiment of the present disclosure.

[0011] FIG. 2 provides a side, cross-sectional view of the dishwashing appliance of FIG. 1.

[0012] FIG. 3 provides a perspective view of a multichannel dispenser assembly in accordance with an exemplary embodiment of the present disclosure.

[0013] FIG. 4 provides a cross sectional view of the multichannel dispenser assembly of FIG. 3 along line 4-4.

[0014] FIG. 5 provides an exploded view of the multichannel dispenser assembly of FIG. 3.

[0015] FIGS. 6A-6B provide alternative embodiments of a seal for a multichannel dispenser assembly in accordance with an exemplary embodiment of the present disclosure.

[0016] FIGS. 7A-7B provide cross-sectional views of the multichannel dispenser assembly of FIG. 3 in different positions.

[0017] FIG. 8 provides a flow diagram of a method for operating a multichannel dispenser assembly in accordance with an exemplary embodiment of the present disclosure.

[0018] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION

[0019] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0020] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). In addition, here and throughout the specification and claims, range limitations may be combined and/or interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0021] Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or

terms, such as “generally,” “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 10 percent margin, i.e., including values within ten percent greater or less than the stated value. In this regard, for example, when used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction, e.g., “generally vertical” includes forming an angle of up to ten degrees in any direction, e.g., clockwise or counterclockwise, with the vertical direction V.

[0022] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” In addition, references to “an embodiment” or “one embodiment” does not necessarily refer to the same embodiment, although it may. Any implementation described herein as “exemplary” or “an embodiment” is not necessarily to be construed as preferred or advantageous over other implementations. Moreover, each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0023] Referring now to the drawings, FIGS. 1 and 2 illustrate a front view of an appliance that may be configured in accordance with aspects of the present disclosure. For the particular embodiment of FIGS. 1 and 2, the appliance is a dishwasher appliance **100**. As shown particularly in FIG. 2, dishwasher appliance **100** may include a cabinet **102** having a tub **104** therein that defines a wash chamber **106** for the receipt of articles for washing. Tub **104** may include a front opening (not shown) and a door **110** for selectively opening and closing wash chamber **106**. As shown, door **110** may be hinged to wash chamber **106** for movement between a closed position (shown in FIGS. 1 and 2) and an open position for loading and unloading articles into and from dishwasher **100**. In addition, as shown, a latch **116** may be used to lock and unlock door **110** for access to wash chamber **106**.

[0024] As shown in FIG. 2, dishwasher appliance **100** may be further equipped with a controller **156** to regulate operation of the dishwasher appliance **100**. Controller **156** may include one or more memory devices and one or more microprocessors, such as general or special purpose microprocessors operable to execute programming instructions or micro-control code associated with a cleaning cycle. The memory may represent random access memory such as DRAM or read only memory such as ROM or FLASH. In an embodiment, the processor executes programming instructions stored in the memory. The memory may be a separate component from the processor or may be included onboard within the processor. Further, controller **156** may be positioned in a variety of locations throughout dishwasher appliance **100**. In the illustrated embodiment, for example, controller **156** may be located within a control panel area **158** of door **110** as shown in FIG. 1. Typically, controller **156** includes a user interface panel/controls **160** through which a user may select various operational features and modes and monitor progress of dishwasher appliance **100**.

[0025] Referring particularly to FIG. 2, dishwasher appliance **100** may further include a fluid reservoir **170** containing a plurality of fluids **178** for washing and/or rinsing the articles within wash chamber **106**. Furthermore, as shown, dishwasher appliance **100** may also include a pump **174** for directing or pumping the plurality of fluids **178** from fluid reservoir **170** and into wash chamber **106**. Pump **174** may selectively pump fluid (e.g., detergent) from fluid storage vessel **178** (e.g., via a diverter, described below) according to certain wash cycle needs of dishwashing appliance **100**. Accordingly, pump **174** may be any suitable pump. In at least one embodiment,

pump **174** is a peristaltic pump. However, it should be understood that the described pump is provided by way of example only and the disclosure is not limited to the examples provided herein. Further, and as will be discussed in more detail herein, dishwasher appliance **100** may include a dispenser assembly **200** for controlling the plurality of fluids **178** flowing from fluid reservoir **170** and into wash chamber **106** along a flow path **172**.

[0026] As shown in FIG. 2, dispenser assembly **200** may be in fluid communication with a dispenser **176** for dispensing one of the plurality of different fluids into wash chamber **106** (e.g., one different fluid at a time). As shown in FIG. 2, fluid reservoir **170**, pump **174**, dispenser **176**, and dispenser assembly **200** may be disposed within door **110**. As such, FIG. 2 depicts fluid reservoir **170**, dispenser assembly **200**, pump **174**, and dispenser **176** as having a particular arrangement along flow path **172**. However, it should be understood that such an arrangement is not limiting, and a variety of arrangements can be implemented in accordance with the present disclosure. For example, pump **174** and dispenser **176** may be arranged in a different order along flow path **172** such that dispenser **176** is disposed adjacent or connected to dispenser assembly **200**.

[0027] Referring now to FIG. 3, a perspective view of dispenser assembly **200** according to an exemplary embodiment of the present disclosure is illustrated. Dispenser assembly **200** may be used with a variety of appliances, such as dishwasher appliance **100** of FIGS. 1 and 2. As shown, dispenser assembly **200** may include a housing **210** having a plurality of inlets (e.g., fluid inlet channels) **216** and an outlet (e.g., fluid outlet channel) **218**. In an embodiment, each of the plurality of inlets **216** may extend from housing **210** in a first direction, whereas outlet **218** may extend from housing **210** in a second direction, the second direction being different from the first direction. As such, in an embodiment, the first and second directions may be generally perpendicular. Further, the plurality of inlets **216** may include any suitable number of inlets, such as four separate inlets as shown. In further embodiments, dispenser assembly **200** may have more or less than four separate inlets.

[0028] Still referring to FIG. 3, housing **210** of dispenser assembly **200** may also have a first housing portion **212** detachably secured to a second housing portion **214**. Further, in an embodiment, as shown, first housing portion **212** and second housing portion **214** may be secured together via one or more fasteners **206**. In some embodiments, fastener(s) **206** include a variety of different fasteners, such as screws, nuts and bolts, or any other suitable fasteners. In addition, according to an embodiment, first housing portion **212** and second housing portion **214** may each include a plurality of projections **202** and **204**, respectively, for securing first housing portion **212** with second housing portion **214**. For example, the plurality of projections **202**, **204** of first and second housing portions **212**, **214** may be aligned and fasteners **206** may extend therethrough.

[0029] Referring now to FIG. 4, a cross-sectional view of dispenser assembly **200** of FIG. 3 along line 4-4 is illustrated. As shown, dispenser assembly **200** may include a rotatable disk **220** positioned within housing **210**. Rotatable disk **220** may include an aperture **222**. Aperture **222** may be selectively aligned with one of the plurality of inlets **216** of housing **210** to provide a flow path **240** between one of the plurality of inlets **216** and outlet **218**.

[0030] A fluid volume **226** may be defined within housing **210**. For example, fluid volume **226** may be defined between first housing portion **312** and second housing portion **214**. In some embodiments, fluid volume **226** is defined between second housing portion **214** and rotatable disk **220**. In such embodiments, rotatable disk **220** includes protrusion **224** extending through second housing portion **214** and beyond housing **210**. Rotatable disk **220** may also include a gauge section **228** formed between protrusion **224** and aperture **222** of rotatable disk **220**. Gauge section **228** may not extend through or beyond second housing portion **214**. In some instances, gauge section **228** is located inside housing **210** and abuts second housing portion **214**. Gauge section **228** may separate aperture **222** of rotatable disk **220** from second housing portion **214** such that fluid volume **226** is defined between second housing portion **214** and rotatable disk **220**. Thus, as shown in FIG. 4, flow path **240** may pass through fluid volume **226**.

[0031] Referring to FIGS. 4 and 5, rotatable disk 220 may include protrusion 224 extending through second housing portion 214 and beyond housing 210. Thus, in some embodiments, protrusion 224 may be mechanically coupled to an actuator 230 for driving rotation of rotatable disk 220 about axis A. Actuator 230 may drive rotatable disk 220 about axis A to align aperture 222 of rotatable disk 220 with one of the plurality of inlets 216. Moreover, protrusion 224 may have any suitable shape, such as a generally cylindrical shape. As shown in FIG. 5, protrusion 224 may have a generally cylindrical shape with a flat surface 225. In such embodiments, flat surface 225 of protrusion 224 may facilitate mechanical coupling with actuator 230 for driving rotation of rotatable disk 220.

[0032] Dispenser assembly 200 may be configured for being fluidly coupled with a fluid reservoir 170 containing a plurality of different fluids 178. Dispenser assembly 200 may be fluidly coupled with fluid reservoir 170 (FIGS. 2 and 4) containing the plurality of fluids 178 via a plurality of conduits 180 coupled to the plurality of inlets 216 of housing 210. Thus, one of the plurality of fluids 178 may flow into fluid volume 226 through one of the plurality of inlets 216 and fill fluid volume 226. Further, outlet 218 of housing 210 may be connected to a pump, such as pump 174 of dishwasher appliance 100 shown in FIG. 1. Accordingly, fluid pumped through dispenser assembly 200 may be provided to dispenser 176 of dishwasher appliance 100 via pump 174. As shown in FIG. 4, a pump line 175 may be connected between outlet 218 of housing 210 and pump 174 of dishwasher appliance 100 for the different fluids 178 to be pumped therethrough. As such, dispenser 176 may be configured to dispense one of the plurality of different fluids 178 into appliance 100 at a time.

[0033] Referring still to FIGS. 4 and 5, dispenser assembly 200 may also include at least one inlet seal 260 within housing 210 to fluidly separate each of the plurality of inlets 216. Dispenser assembly 200 may also include at least one housing seal 270. The housing seal 270 may be arranged between first housing portion 212 and second housing portion 214 and/or protrusion 224 of rotatable disk 220 and second housing portion 214. A first housing seal 271 may be arranged between first and second housing portions 212, 214, and a second housing seal 272 may be arranged between protrusion 224 and second housing portion 214. Further, second housing portion 214 may define an opening 215 for protrusion 224 to extend therethrough.

[0034] Additionally or alternatively, as shown generally in FIGS. 4 and 5, dispenser assembly 200 may include spaces and/or gaps between certain features to facilitate operation thereof. For example, as shown in FIG. 4, a plurality of recesses 208 may be formed into at least one of first housing portion 212 and second housing portion 214. More specifically, one or more of recesses 208 may be formed between first housing portion 212 and rotating disk 220. Additionally or alternatively, one or more of recesses 208 may be formed between first housing portion 212 and second housing portion 214. Furthermore, one or more of recesses 208 may be formed between rotating disk 220 and second housing portion 214. Accordingly, the at least one inlet seal 260 and/or the at least one housing seal 270 may fit within the one or more recesses 208, such as the one or more recesses formed into at least one of first housing portion 212 or second housing portion 214.

[0035] For example, the at least one inlet seal 260 may fit within the one or more recesses 208 formed between first housing portion 212 and rotating disk 220. The at least one housing seal 270 may fit within the one or more recesses 208 formed between first housing portion 212 and second housing portion 214 or the one or more recesses 208 formed between second housing portion 214 and rotating disk 220. First housing seal 271 may fit within recess 208 formed between first and second housing portions 212, 214, while second housing seal 272 may fit within recess 208 between second housing portion 214 and rotating disk 220. In some embodiments, as shown in FIG. 5, inlet seal(s) 260 and/or housing seal(s) 270 may include one or more O-rings.

[0036] Referring to FIGS. 6A-6B, further embodiments of inlet seal(s) 360 of dispenser assembly 200 are illustrated according to the present disclosure. For instance, FIG. 6A provides a multi-seal

arrangement **300A** of inlet seal **260**. As shown, multi-seal arrangement **300A** may include a plurality of inlet seals **302A** with one of the plurality of inlet seals **302A** being arranged with each of the plurality of inlets **216**. The plurality of inlet seals **302A** may include a first inlet seal **304A** arranged with one of the plurality of inlets **216**. Accordingly, the plurality of inlet seals **302A** may include a number of inlet seals that is equivalent to a number of each inlet of the plurality of inlets **216**.

[0037] FIG. **6B** illustrates a segmented-ring seal **300B** of inlet seal **260**. Segmented-ring seal **300B** may include a single inlet seal that defines a plurality of openings **302B**. One of the plurality of openings **302B** may be arranged with each of the plurality of inlets **216**. As shown in FIG. **6B**, the second inlet seal includes a ring **408** and a plurality of cross-connecting segments **410**, thereby defining the plurality of openings **302B**. The plurality of openings **302B** may include a first opening **304B** arranged with one of the plurality of inlets **216**. Accordingly, the plurality of openings **302B** may include a number of openings that is equivalent to a number of inlets of the plurality of inlets **216** (e.g., four).

[0038] Referring now to FIGS. **7A-7B**, cross-sectional views of different positions of the dispenser assembly **200** of FIG. **3** are illustrated according to the present disclosure. The cross-sectional views depicted in FIGS. **7A-7B** are described herein with further reference to FIGS. **1-5** and **6A-6B**. Further, in FIGS. **7A-7B**, the cross-sectional views are taken along the line **4-4** shown in FIG. **4**. As such, dispenser assembly **200** may include rotatable disk **220** with aperture **222** and inlet seal **260** disposed between rotatable disk **220** and first housing portion **212**. Referring particularly to FIG. **7A**, aperture **222** of rotatable disk **220** may be in a first position **401**. In the first position **401**, aperture **222** is selectively aligned with an inlet (e.g., a first inlet or first fluid inlet channel) of the plurality of inlets **216** to provide a flow path **240** between the first fluid inlet channel of the plurality of inlets **216** and outlet **218** of housing **210**. In the first position **401**, the remaining inlets of the plurality of inlets **216** are blocked by rotatable disk **220** while allowing the first fluid inlet channel to facilitate releasing one of the plurality of fluids **178** contained within fluid reservoir **170**. Actuator **230** of dispenser assembly **200** may be configured to drive rotation of rotatable disk **220** about axis **A**. As such, rotatable disk **220** may be rotated in a rotational direction **404** such that aperture **222** is no longer in the first position **401**.

[0039] As shown in FIG. **7B**, rotatable disk **220** may be rotated such that aperture **222** is in a second position **402**. In the second position **402**, aperture **222** is selectively aligned with a different inlet (e.g., a second inlet or second fluid inlet channel) of the plurality of inlets **216** to provide a flow path **240** between the second fluid inlet channel of the plurality of inlets **216** and outlet **218** of housing **210**. As will be described in more detail below, the second fluid inlet channel may be an open-air inlet. In detail, no fluid from the plurality of fluids **178** from fluid reservoir **170** is connected to the second fluid inlet channel. Furthermore, it should be understood that any of the plurality of inlets **216** may be referred to as the second fluid inlet channel or open-air channel. Thus, any of the plurality of inlets **216** may be an open-air inlet channel. When rotatable disk **220** is in the second position **402**, air (e.g., ambient air) may be pumped through the second fluid inlet channel and into, e.g., flow path **172**, pump line **175**, etc. As such, flow path **172** may be purged of any residual fluid remaining in flow path **172**, pump line **172**, or the like.

[0040] Now that the general descriptions of an exemplary appliance have been described in detail, a method **500** of operating an appliance (e.g., dishwasher appliance **100**) will be described in detail. Although the discussion below refers to the exemplary method **500** of operating dishwasher appliance **100**, one skilled in the art will appreciate that the exemplary method **500** is applicable to any suitable domestic appliance capable of performing a dispensing operation (e.g., such as a washing machine appliance, etc.). In exemplary embodiments, the various method steps as disclosed herein may be performed by controller **156** and/or a separate, dedicated controller. FIG. **8** provides a flow chart illustrating a method of operating a dishwasher appliance. Hereinafter, method **500** will be described with specific reference to FIG. **8**.

[0041] At step **502**, method **500** may include receiving a signal to dispense a predetermined volume or amount of fluid from at least one fluid storage vessel via a fluid diverter. As discussed above, fluid storage vessels or bulk tanks may be included within or fluidly attached to the appliance (e.g., dishwasher appliance **100**). Further, the dispenser assembly (e.g., dispenser assembly **200**) may be in fluid communication with each of the fluid storage vessels as well as a chamber of the appliance. The signal may include a user input to initiate a washing operation, such as a dishwashing operation. Accordingly, the signal may be received via a user interface of the appliance, via a remote input from a remote device, via an automatic or preprogrammed request, or the like.

[0042] At step **504**, method **500** may include selecting a first fluid inlet channel in response to receiving the signal. The first fluid inlet channel may be one of a plurality of fluid inlet channels (e.g., inlets **218**). The first fluid inlet channel may be in fluid communication with a first bulk storage vessel or tank, such as a detergent tank. In selecting the first fluid inlet channel, method **500** may include rotating, via an actuator (e.g., motor), a rotatable disk (e.g., rotatable disk **220**) to a first position. As discussed above, the first position may align an aperture (e.g., through hole or aperture **222**) with the first fluid inlet channel. The signal may thus include a command to operate the actuator (or motor) to rotate the rotatable disk to a predetermined position (e.g., first position) or through a predetermined rotational arc.

[0043] At step **506**, method **500** may include initiating a pump to pump the predetermined volume of fluid from the at least one fluid storage vessel. In detail, after selecting the first fluid inlet channel, method **500** may direct the pump (e.g., pump **174**) at a predetermined level to supply the fluid from the at least one fluid storage vessel to the fluid diverter and subsequently to the wash chamber of the appliance. The signal to dispense the predetermined volume of fluid may include the predetermined volume of fluid, for instance, in milliliters, ounces, or the like. For instance, according to a selected operation (e.g., cleaning cycle), the pump may operate according to a predetermined length of time to accurately dispense the predetermined volume of fluid. In additional or alternative embodiments, the pump may be operated according to a predetermined pattern, such as a number of rotations, iterations, cycles, or the like (e.g., in the case of a peristaltic pump). Accordingly, the predetermined pattern or length of time may be based on the predetermined volume of fluid. It should be understood that any suitable pump may be incorporated and any suitable measurement of the fluid may be used in specific embodiments.

[0044] At step **508**, method **500** may include selecting a second fluid inlet channel of the plurality of fluid inlet channels after pumping the predetermined volume of fluid. The second fluid inlet channel may be an open-air inlet channel. As described above, the plurality of fluid inlet channels may include at least one fluid inlet channel (e.g., the second fluid inlet channel) that is not connected to any tank, vessel, or storage container. As such, air (e.g., ambient air) may be pumped through the second fluid inlet channel.

[0045] In selecting the second fluid inlet channel, method **500** may include rotating, via the actuator (e.g., motor), the rotatable disk to a second position. As discussed above, the second position may align the aperture (e.g., through hole or aperture **222**) with the second fluid inlet channel (e.g., the open-air inlet channel). The signal may thus include a command to operate the actuator (or motor) to rotate the rotatable disk to a predetermined position (e.g., second position) or through a predetermined rotational arc.

[0046] Method **500** may further include initiating the pump to purge the fluid from the pump line after selecting the second fluid inlet channel. For instance, after aligning the aperture or through hole with the second fluid inlet channel (e.g., the open-air channel), the pump may be activated to draw air through the second fluid inlet channel and into the fluid diverter. As mentioned above, one or more flow lines, such as the pump line, may be provided downstream from the fluid diverter. The one or more flow lines may further include an inlet line into the wash chamber of the appliance. By operating the pump after selecting the open-air channel, residual fluid within the flow lines (e.g., the pump line, the inlet line, etc.) may be purged or flushed into the wash chamber.

Advantageously, method **500** may ensure that an entire determined dosage of the fluid (e.g., detergent) is supplied to the wash chamber. Additionally or alternatively, method **500** may ensure that two or more fluids are not inadvertently mixed within the flow lines by purging the flow lines after supplying a fluid therethrough.

[0047] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A dispenser assembly for an appliance, the dispenser assembly comprising: a fluid diverter comprising a plurality of fluid inlet channels, a fluid outlet channel, and a diverter input shaft; a pump fluidly coupled to the fluid diverter for selectively urging a flow of fluid from the fluid diverter into the appliance; at least one fluid storage vessel configured to store a fluid, the at least one fluid storage vessel fluidly connected with the fluid diverter via a first fluid inlet channel of the plurality of fluid inlet channels; and a controller operably connected with the fluid diverter and the pump, the controller being configured to perform an operation, the operation comprising: receiving a signal to dispense a predetermined volume of the fluid from the at least one fluid storage vessel via the fluid diverter; selecting the first fluid inlet channel in response to receiving the signal; initiating the pump to pump the predetermined volume of the fluid from the at least one fluid storage vessel after selecting the first fluid inlet channel; and selecting a second fluid inlet channel of the plurality of fluid inlet channels after pumping the predetermined volume of the fluid, wherein the second fluid inlet channel is an open-air inlet channel.
2. The dispenser assembly of claim 1, wherein the fluid diverter further comprises: a housing, wherein each of the plurality of fluid inlet channels and the fluid outlet channel is formed through the housing; a rotatable disk positioned within the housing, the rotatable disk comprising an aperture for being selectively aligned with one of the plurality of fluid inlet channels to provide a flow path between one of the plurality of fluid inlet channels and the fluid outlet channel; and an actuator mechanically coupled to the rotatable disk for driving rotation of the rotatable disk about an axis to align the aperture of the rotatable disk with one of the plurality of fluid inlet channels.
3. The dispenser assembly of claim 2, wherein selecting the first fluid inlet channel comprises: rotating, via the actuator, the rotatable disk to a first position at which the aperture is aligned with the first fluid inlet channel.
4. The dispenser assembly of claim 3, wherein selecting the second fluid inlet channel comprises: rotating, via the actuator, the rotatable disk to a second position at which the aperture is aligned with the open-air inlet channel.
5. The dispenser assembly of claim 1, further comprising: a pump line fluidly coupled between the fluid outlet channel and the pump.
6. The dispenser assembly of claim 5, wherein the operation further comprises: initiating the pump to purge the fluid from the pump line after selecting the second fluid inlet channel.
7. The dispenser assembly of claim 1, wherein initiating the pump to pump the predetermined volume of fluid from the at least one fluid storage vessel comprises: activating the pump according to a predetermined pattern.
8. The dispenser assembly of claim 7, wherein the predetermined pattern is based on the predetermined volume of the fluid.

9. The dispenser assembly of claim 1, wherein the appliance is a dishwasher appliance comprising a tub and a door moveable between an open position and a closed position, wherein the dispenser assembly is provided within the door.

10. A method of operating a dispenser assembly of an appliance, the dispenser assembly comprising a fluid diverter comprising a plurality of fluid inlet channels and a fluid outlet channel, a pump fluidly coupled to the fluid diverter, and at least one fluid storage vessel, the method comprising: receiving a signal to dispense a predetermined volume of a fluid from the at least one fluid storage vessel via the fluid diverter; selecting a first fluid inlet channel in response to receiving the signal, the first fluid inlet channel being fluidly connected with the at least one fluid storage vessel; initiating the pump to pump the predetermined volume of the fluid from the at least one fluid storage vessel after selecting the first fluid inlet channel; and selecting a second fluid inlet channel of the plurality of fluid inlet channels after pumping the predetermined volume of the fluid, wherein the second fluid inlet channel is an open-air inlet channel.

11. The method of claim 10, wherein the fluid diverter further comprises: a housing, wherein each of the plurality of fluid inlet channels and the fluid outlet channel is formed through the housing; a rotatable disk positioned within the housing, the rotatable disk comprising an aperture for being selectively aligned with one of the plurality of fluid inlet channels to provide a flow path between one of the plurality of fluid inlet channels and the fluid outlet channel; and an actuator mechanically coupled to the rotatable disk for driving rotation of the rotatable disk about an axis to align the aperture of the rotatable disk with one of the plurality of fluid inlet channels.

12. The method of claim 11, wherein selecting the first fluid inlet channel comprises: rotating, via the actuator, the rotatable disk to a first position at which the aperture is aligned with the first fluid inlet channel.

13. The method of claim 12, wherein selecting the second fluid inlet channel comprises: rotating, via the actuator, the rotatable disk to a second position at which the aperture is aligned with the open-air inlet channel.

14. The method of claim 10, further comprising: a pump line fluidly coupled between the fluid outlet channel and the pump.

15. The method of claim 14, further comprising: initiating the pump to purge the fluid from the pump line after selecting the second fluid inlet channel.

16. The method of claim 10, wherein initiating the pump to pump the predetermined volume of fluid from the at least one fluid storage vessel comprises: activating the pump according to a predetermined pattern.

17. The method of claim 16, wherein the predetermined pattern is based on the predetermined volume of the fluid.

18. The method of claim 10, wherein the appliance is a dishwasher appliance comprising a tub and a door moveable between an open position and a closed position, wherein the dispenser assembly is provided within the door.
